

HTML & CSS

Thierry Sans

Client Side (a.k.a Frontend)

HTML	Content
CSS	Presentation
Javascript	Processing

HTML - defining the content

HTML : Hyper-Text Markup Language



HTML 1	invented by Tom Berners-Lee (1991)
HTML 2	first standard from W3C (1995)
HTML 4	separation between content and presentation CSS (1997)
HTML 5	multimedia (2008)

Skeleton of an HTML document

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8"/>
    <title>My first page</title>
  </head>
  <body>
    <!-- It's a joke -->
    <h1 class="header">CSCC09</h1>
    <p>Check the
      <a href="https://thierrysans.me/CSCC09/">Website</a>
    </p>
  </body>
</html>
```



Metadata is stored in **the header**

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8"/>
    <title>My first page</title>
  </head>
  <body>
    <!-- It's a joke -->
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    </p>
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```

Content is stored in **the body**

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<!DOCTYPE html>
<html>
  <head>
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      <a href="https://thierrysans.me/CSCC09/">Website</a>
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Content is stored in **the body**

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<!DOCTYPE html>
<html>
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    <h1 class="header">CSCC09</h1>
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      <a href="https://thierrysans.me/CSCC09/">Website</a>
    </p>
  </body>
</html>
```

HTML Element

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8"/>
    <title>My first page</title>
  </head>
  <body>
    <!-- It's a joke -->
    <b1 class="header">CSCC09</b1>
    <p>Check the
      <a href="https://thierrysans.me/CSCC09/">Website</a>
    </p>
  </body>
</html>
```

Attributes (key/values pairs) in the opening tag

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8"/>
    <title>My first page</title>
  </head>
  <body>
    <!-- It's a joke -->
    <h1 class="header">CSCC09</h1>
    <p>Check the
      <a href="https://thierrysans.me/CSCC09/">Website</a>
    </p>
  </body>
</html>
```

Selection of tags

<code><h1></code> <code><h2></code> <code><h3></code>	headers (<code><h1></code> is the biggest by convention)
<code><p></code>	delimitates a paragraph
<code></code>	image
<code><a></code>	specifies an anchor (typically an hyperlink)
<code></code>	unordered list
<code></code>	list item
<code><div></code>	delimitates a block
<code></code>	group in-line document (use mainly for style)

Differences between XHTML and HTML

At first, browsers were quite forgiving about missing/mismatched tags in HTML

- Different rendering across browsers

HTML became XHTML (aka HTML 4 and 5)

- ✓ Homogeneity across browsers (for the most part)
- ✓ Easier to parse in Javascript

Validate your (X)HTML

Check out W3C Markup Validation Service:

<http://validator.w3.org/>

Or HTML Tidy (command line tool):

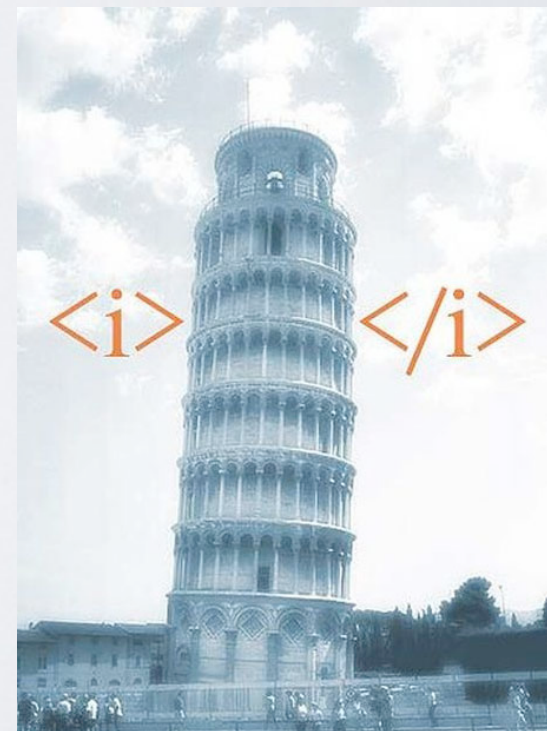
<http://www.html-tidy.org/>

Differences between XHTML and HTML

The styling disappeared in HTML 4 to become CSS

➔ Elements related to styling became deprecated

`<center>` `<b>` `<i>` `<br>` `<font>` `<frameset>` `<frame>` ...



Controversies among web developers

To use or not to use tables

<http://programmers.stackexchange.com/questions/277778/why-are-people-making-tables-with-divs>

CSS - defining the style

Is it really used and useful?

Let's look at your favourite websites without CSS

Why CSS?

Multiple pages - one style

Multiple platforms - different layouts

Multiple user abilities - custom layout

CSS format

```
selector {  
    property: value;  
    property: value;  
    property: value;  
}
```

CSS: Inline, embedded or separate file?

Inline: `style` attribute

```
<p style="background: blue; color: white;"> ...
```

Embedded: specified in the header (`<head>`)

```
<style TYPE="text/css">  
    p{background: blue; color: white;}  
</style>
```

Separate file: file locator in the header (`<head>`)

```
<link rel="stylesheet" type="text/css" href="style.css"/>
```

Classes

Classes are attributes of HTML elements for which we want to define **common properties**

```
<div class="special">...</div>
```

Define the same style for all elements of the same class

```
.special {  
    margin-top: 10px;  
    font-family: "Helvetica", "Arial";  
}
```

IDs

IDs are attributes of HTML elements for which we want to **identify uniquely**

```
<div id="sale">...</div>
```

Define the style for the element with specific ID

```
#sale {  
    padding: 20px;  
    color: #000000;  
}
```

The CSS Box Model

Let's look at

<https://css-tricks.com/the-css-box-model/>

Positioning and Floating (aka “The Big Headache”)

Let's look at

<https://css-tricks.com/almanac/properties/p/position/>

Flexbox (my favorite)

<https://css-tricks.com/snippets/css/a-guide-to-flexbox/>

CSS Grid (the latest addition)

<https://css-tricks.com/snippets/css/complete-guide-grid/>